

Lecture 15

Deterministic computation

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(Edited from the lecture notes by Brian J. Reich, NC State)



Brief recap of Chapter 1 and 2

- ▶ Probability essentials (PMF, PDF, marginal and conditional distributions)
- ▶ Bayes' theorem and Bayes versus frequentist paradigm
- ▶ Posterior summarization and posterior predictive distribution
- ▶ Conjugate priors
- ▶ Prior elicitation
- ▶ Improper priors and objective priors (Jeffreys' prior)

Bayesian computing

- ▶ Given the prior and data, the posterior is fixed and a Bayesian analysis boils down to summarizing the posterior.
- ▶ We need point estimates, credible sets, etc.
- ▶ Summarizing a P -dimensional posterior distribution is challenging for large P .
- ▶ In the 80's, Bayesian computing was unable to do this for more than a few parameters.
- ▶ In the 90's, new algorithms were developed that revolutionized Bayesian statistics.
- ▶ Understanding these algorithms is obviously important.

Sampling-Based Approaches to Calculating Marginal Densities

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Stochastic substitution, the Gibbs sampler, and the sampling–importance–resampling algorithm can be viewed as three alternative sampling- (or Monte Carlo-) based approaches to the calculation of numerical estimates of marginal probability distributions. The three approaches will be reviewed, compared, and contrasted in relation to various joint probability structures frequently encountered in applications. In particular, the relevance of the approaches to calculating Bayesian posterior densities for a variety of structured models will be discussed and illustrated.

KEY WORDS: Conditional probability structure; Data augmentation; Gibbs sampler; Hierarchical models; Importance sampling; Missing data; Monte Carlo sampling; Posterior distributions; Stochastic substitution; Variance components.

Approaches to Bayesian computing

Some approaches to dealing with complicated joint posteriors:

- ▶ Just use a point estimate (e.g., MAP), ignore uncertainty
- ▶ Approximate the posterior as Gaussian
- ▶ Numerical integration
- ▶ Markov Chain Monte Carlo (MCMC) sampling

Outline of Chapter 3

- ▶ Deterministic methods
 - ▶ MAP estimation
 - ▶ Numerical integration
 - ▶ Bayesian Central Limit Theorem
- ▶ MCMC algorithms
 - ▶ Gibbs sampling
 - ▶ Metropolis-Hastings sampling
- ▶ Just Another Gibbs Sampler (JAGS)
- ▶ Diagnostic and improving convergence
 - ▶ Setting initial values
 - ▶ Convergence diagnostics
 - ▶ Improving convergence
 - ▶ Dealing with large datasets

MAP estimation

- ▶ Sometimes you don't need an entire posterior distribution and a single point estimate works fine.
- ▶ Example: prediction in machine learning.
- ▶ The Maximum a Posteriori (MAP) estimate is the posterior mode

$$\hat{\theta}_{MAP} = \operatorname{argmax}_{\theta} p(\theta|\mathbf{Y}) = \operatorname{argmax}_{\theta} \log[f(\mathbf{Y}|\theta)] + \log[\pi(\theta)].$$

- ▶ This is similar to the maximum likelihood estimation but includes the prior.

Univariate example 1

Say $Y|\theta \sim \text{Poisson}(N\theta)$ and $\theta \sim \text{Gamma}(a, b)$, find $\hat{\theta}_{MAP}$.

- ▶ The likelihood is $f(Y|\theta) \propto \exp[-N\theta]\theta^Y$.
- ▶ The log-likelihood is $\ell(\theta|Y) = \log[f(Y|\theta)] = c_f - N\theta + Y \log(\theta)$.
- ▶ The prior is $\pi(\theta) \propto \theta^{a-1} \exp[-b\theta]$.
- ▶ The log-prior is $\log[\pi(\theta)] = c_\pi + (a-1) \log(\theta) - b\theta$.
- ▶ Thus, the MAP estimate is

$$\hat{\theta}_{MAP} = \underset{\theta}{\operatorname{argmax}} p(\theta|Y) = \underset{\theta}{\operatorname{argmax}} \{ \log[f(Y|\theta)] + \log[\pi(\theta)] \}.$$

Univariate example 1 (contd.)

Say $Y|\theta \sim \text{Poisson}(N\theta)$ and $\theta \sim \text{Gamma}(a, b)$, find $\hat{\theta}_{MAP}$.

- ▶ Thus, the MAP estimate is

$$\hat{\theta}_{MAP} = \underset{\theta}{\operatorname{argmax}} \{ -(N + b)\theta + (Y + a - 1) \log(\theta) \}.$$

- ▶ Taking the derivative and setting to zero gives

$$-(N + b) + (Y + a - 1) \frac{1}{\theta} = 0.$$

- ▶ The solution is

$$\theta = \frac{Y + a - 1}{N + b}.$$

Univariate example 2

Say $Y|\theta \sim \text{Binomial}(N, \theta)$ and $\theta \sim \text{Beta}(a, b)$, find $\hat{\theta}_{MAP}$.

- ▶ The likelihood is $f(Y|\theta) \propto \theta^Y(1 - \theta)^{N-Y}$.
- ▶ The log-likelihood is $\ell(\theta|Y) = \log[f(Y|\theta)] = c_f + Y \log(\theta) + (N - Y) \log(1 - \theta)$.
- ▶ The prior is $\pi(\theta) \propto \theta^{a-1}(1 - \theta)^{b-1}$.
- ▶ The log-prior is $\log[\pi(\theta)] = c_\pi + (a - 1) \log(\theta) + (b - 1) \log(1 - \theta)$.
- ▶ Thus, the MAP estimate is

$$\hat{\theta}_{MAP} = \underset{\theta}{\operatorname{argmax}} p(\theta|Y) = \underset{\theta}{\operatorname{argmax}} \{ \log[f(Y|\theta)] + \log[\pi(\theta)] \}.$$

Univariate example 2 (contd.)

Say $Y|\theta \sim \text{Binomial}(N, \theta)$ and $\theta \sim \text{Beta}(a, b)$, find $\hat{\theta}_{MAP}$.

- ▶ Thus, the MAP estimate is

$$\hat{\theta}_{MAP} = \underset{\theta}{\operatorname{argmax}} \{ (Y + a - 1) \log(\theta) + (N - Y + b - 1) \log(1 - \theta) \}.$$

- ▶ Taking the derivative and setting to zero gives

$$\frac{Y + a - 1}{\theta} - \frac{N - Y + b - 1}{1 - \theta} = 0.$$

- ▶ The solution is

$$\theta = \frac{Y + a - 1}{Y + a - 1 + N - Y + b - 1} = \frac{Y + a - 1}{N + a + b - 2}.$$

Multivariate example

Say $\mathbf{Y}|\boldsymbol{\mu} \sim \text{Normal}_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\boldsymbol{\mu} \sim \text{Normal}_p(\boldsymbol{\mu}_0, m^{-1}\boldsymbol{\Sigma})$. Here, $\boldsymbol{\Sigma}$ is known.

- ▶ The likelihood is $f(\mathbf{Y}|\boldsymbol{\mu}) = |2\pi\boldsymbol{\Sigma}|^{-1/2} \exp \left[-\frac{1}{2}(\mathbf{Y} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu}) \right]$.
- ▶ The log-likelihood is $\ell(\boldsymbol{\mu}|\mathbf{Y}) = \log[f(\mathbf{Y}|\boldsymbol{\mu})] = c_f - \frac{1}{2}(\mathbf{Y} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$.
- ▶ The prior is $\pi(\boldsymbol{\mu}) \propto \exp \left[-\frac{m}{2}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \boldsymbol{\mu}_0) \right]$.
- ▶ The log-prior is $\log[\pi(\boldsymbol{\mu})] = c_\pi - \frac{m}{2}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)$.
- ▶ Thus, the MAP estimate is

$$\hat{\boldsymbol{\mu}}_{MAP} = \underset{\boldsymbol{\mu}}{\operatorname{argmax}} p(\boldsymbol{\mu}|\mathbf{Y}) = \underset{\boldsymbol{\mu}}{\operatorname{argmax}} \{ \log[f(\mathbf{Y}|\boldsymbol{\mu})] + \log[\pi(\boldsymbol{\mu})] \}.$$

Multivariate example (contd.)

Say $\mathbf{Y}|\mu \sim \text{Normal}_P(\mu, \Sigma)$ and $\mu \sim \text{Normal}_P(\mu_0, m^{-1}\Sigma)$. Here, Σ is known.

- ▶ Thus, the MAP estimate is

$$\hat{\mu}_{MAP} = \underset{\mu}{\operatorname{argmax}} \left\{ -\frac{1}{2}(\mathbf{Y} - \mu)' \Sigma^{-1}(\mathbf{Y} - \mu) - \frac{m}{2}(\mu - \mu_0)' \Sigma^{-1}(\mu - \mu_0) \right\}.$$

- ▶ Taking the multivariate derivative and setting to zero gives

$$(1 + m)\Sigma^{-1}\mu - \Sigma^{-1}(\mathbf{Y} + m\mu_0) = \mathbf{0}.$$

- ▶ The solution is

$$\mu = \frac{\mathbf{Y} + m\mu_0}{1 + m}.$$

Bayesian central limit theorem

- ▶ Another simplification is to approximate the posterior as Gaussian.
- ▶ Bernstein-Von Mises Theorem: As the sample size grows the posterior doesn't depend on the prior.
- ▶ Frequentist result: As the sample size grows the likelihood function is approximately normal.
- ▶ Bayesian CLT: For large n and some other conditions, $\theta | \mathbf{Y}_1, \dots, \mathbf{Y}_n \approx \text{Normal}$.

Bayesian central limit theorem

- ▶ Bayesian CLT: For large n and some other conditions

$$\boldsymbol{\theta} | \mathbf{Y}_1, \dots, \mathbf{Y}_n \sim \text{Normal}[\hat{\boldsymbol{\theta}}_{MAP}, \mathbf{H}(\hat{\boldsymbol{\theta}}_{MAP})^{-1}].$$

- ▶ $\mathbf{H}$ is the negative hessian matrix.
- ▶ The (j, k) element of $\mathbf{H}$ is

$$-\frac{\partial^2}{\partial \theta_j \partial \theta_k} \log[p(\boldsymbol{\theta} | \mathbf{Y})]$$

evaluated at $\hat{\boldsymbol{\theta}}_{MAP}$.

- ▶ We have marginal and conditional means, standard deviations and intervals for the normal distribution.

Univariate example

Say $Y|\theta \sim \text{Binomial}(n, \theta)$ and $\theta \sim \text{Beta}(a = 0.5, b = 0.5)$, find the Gaussian approximation for $p(\theta|\mathbf{Y})$.

- ▶ We have seen that

$$\theta = \frac{Y + a - 1}{n + a + b - 2} = \frac{Y - 0.5}{n - 1}.$$

- ▶ After some boring algebra,

$$H(\theta) = \frac{Y + a - 1}{\theta^2} + \frac{n - Y + b - 1}{(1 - \theta)^2}.$$

- ▶ Thus,

$$\theta|Y \sim \text{Normal}[\hat{\theta}_{MAP}, H(\hat{\theta}_{MAP})^{-1}]$$

Illustration of the Bayesian CLT

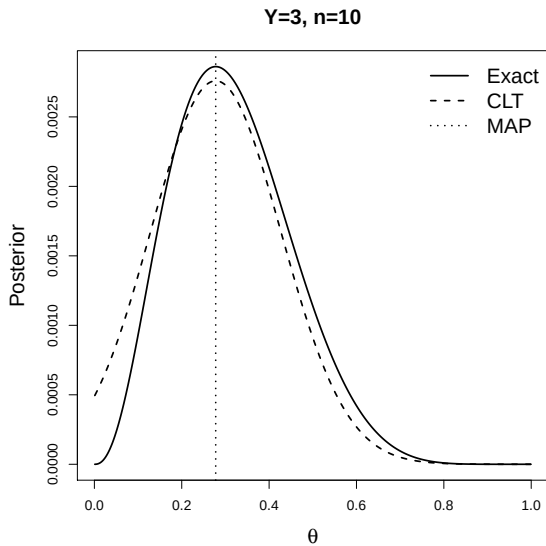


Illustration of the Bayesian CLT

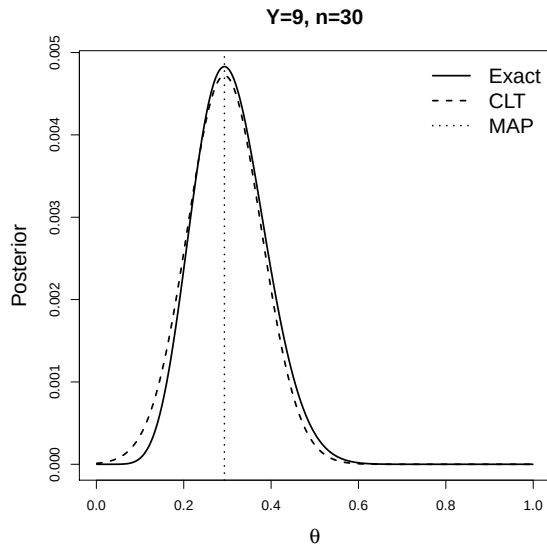
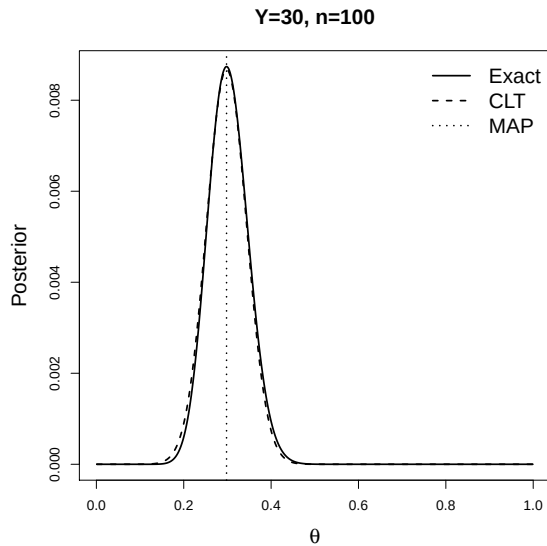


Illustration of the Bayesian CLT



Bayesian central limit theorem

- ▶ For large datasets with a small number of parameters evoking the Bayes CLT is probably the best approach.
- ▶ The approximate posterior can be computed using standard software (e.g., `glm` in R).
- ▶ The numerical values (e.g., intervals) will equal the frequentist values, but the interpretation remains Bayesian.
- ▶ Why not just do a frequentist analysis? Well, why not just do a Bayesian analysis?

Numerical integration

- ▶ Many posterior summaries of interest are integrals over the posterior.
- ▶ Ex: $E(\theta_j|\mathbf{Y}) = \int \theta_j p(\theta) d\theta$.
- ▶ Ex: $V(\theta_j|\mathbf{Y}) = \int [\theta_j - E(\theta|\mathbf{Y})]^2 p(\theta) d\theta$.
- ▶ These are p dimensional integrals that we usually can't solve analytically
- ▶ A grid approximation is a crude approach.
- ▶ Gaussian quadrature is better.

Numerical integration

- ▶ Numerical integration is only feasible for small p .
- ▶ The Iteratively Nested Laplace Approximation (INLA) is an even more sophisticated method.
- ▶ INLA combines Gaussian approximations with numerical integration.
- ▶ This works well if most of the parameters are approximately normal and only a few are non-Gaussian and require numerical integration.

Revolution in high-dimensional Bayesian statistics (JRSS-B, 2009)

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Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations

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[Read before The Royal Statistical Society at a meeting organized by the Research Section on Wednesday, October 15th, 2008, Professor I. L. Dryden in the Chair]

Summary. Structured additive regression models are perhaps the most commonly used class of models in statistical applications. It includes, among others, (generalized) linear models, (generalized) additive models, smoothing spline models, state space models, semiparametric regression, spatial and spatiotemporal models, log-Gaussian Cox processes and geostatistical and geosadditive models. We consider approximate Bayesian inference in a popular subset of structured additive regression models, *latent Gaussian models*, where the latent field is Gaussian, controlled by a few hyperparameters and with non-Gaussian response variables. The posterior marginals are not available in closed form owing to the non-Gaussian response variables. For such models, Markov chain Monte Carlo methods can be implemented, but they are not without problems, in terms of both convergence and computational time. In some practical applications, the extent of these problems is such that Markov chain Monte Carlo sampling is simply not an appropriate tool for routine analysis. We show that, by using an integrated nested Laplace approximation and its simplified version, we can directly compute very accurate approximations to the posterior marginals. The main benefit of these approximations is computational: where Markov chain Monte Carlo algorithms need hours or days to run, our approximations provide more precise estimates in seconds or minutes. Another advantage with our approach is its generality, which makes it possible to perform Bayesian analysis in an automatic, streamlined way, and to compute model comparison criteria and various predictive measures so that models can be compared and the model under study can be challenged.

Thank you!